

Process Characterization Plan

**A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) —
PCP-007**

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available; AMV, analytical method validation; BLA, Biologics

License Application; CQA, critical quality attribute; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PD, Process Development; qPCR, quantitative polymerase chain reaction; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure.

1 Purpose and scope

Cation exchange chromatography is the aggregate polishing step of the A-Mab drug substance process, operated in bind and elute mode on a strong cation exchange resin. On the resin the antibody is captured at low conductivity, weakly bound impurities are removed in the load and in the wash, then a step change in buffer pH recovers the antibody into the eluate pool. Collection of the eluate stops at a fixed point on the descending side of the ultraviolet peak, which leaves the late eluting aggregate on the column. That point is the main lever on pool purity. Pool purity cannot be recovered downstream, because no later step of the purification train reduces aggregate. The aggregate content of the cation exchange pool therefore sets the aggregate content of the drug substance.

The commercial process is being transferred from Novacyte Biologics, Cambridge, MA to Novacyte Biologics, Grafton, WI, under the transfer plan PTP-001. That transfer requires operating ranges for this step that are supported by characterization data generated on a qualified model of the commercial column. The commercial process runs a 15,000 L production bioreactor, from which the chromatography columns are sized. Although the ranges carried over from development have served every clinical campaign, none of them has been challenged across the width now proposed.

The studies that will establish those ranges are stated in this plan, which covers the 5 parameters listed in Table 6.1, the 4 quality attributes listed in Table 4.1, the acceptance criteria in Table 7.1 and the decision rules that will be applied to the results. The regulatory basis for those studies is the enhanced development approach of ICH Q8 and ICH Q11, the risk framework of ICH Q9 and the Stage 1 process design expectations of the lifecycle approach to process validation (International Council for Harmonisation 2009, 2012, 2023b; U.S. Food and Drug Administration 2011), while the methodology follows the A-Mab case study (CMC Biotech Working Group 2009) and the master plan for this campaign (PCMP-001).

The scope of this plan runs from the load of the neutralized viral inactivation pool to the end of eluate collection. Column packing, resin lifetime and sanitization are left to SOP-2008. Viral clearance across this step is not studied here, for the reason given in §4.2, while comparability of the drug substance itself is covered by the transfer plan (PTP-001).

2 Objectives

This plan has five objectives.

- The scale-down model of the step will be qualified against the commercial column before any characterization run is executed (§5.1).
- The effects of protein load, load and wash conductivity, elution buffer pH and elution stop collect on pool aggregate, pool host cell protein and step yield will be quantified across the characterization ranges given in Table 6.1 (§6.2).
- A response-surface model will be fitted to each of the 3 responses, from which the region of the characterized space that meets every pool criterion in Table 7.1 will be defined (§6.3, §7).
- A proven acceptable range within that region will be determined for each governed attribute against each response-surface parameter, both at the set-points of the other parameters and under normal operating range variation (§8).
- The remaining parameter, elution flow rate, will be assessed univariately (§6.4), after which every parameter of the step will be classified under SOP-4001 as an input to the control strategy (§7).

The five objectives share one deliverable, the process characterization report (PCR-007). The contents of that report are described in §11.

3 Related documents

The documents that scope this plan and receive the report are listed in Table 3.1, which names the parent documents, the paired report and the master report the study rolls up into.

Table 3.1: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-007	Process Characterization Report (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and validated methods that govern execution of the step are listed in Table 3.2, with each one cited in the section that relies on it.

Table 3.2: Controlled procedures and validated analytical methods for this step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation

Reference	Title	Type
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The step follows low-pH viral inactivation and precedes anion exchange, taking the neutralized inactivation pool as the load after adjustment to the load conductivity given in Table 6.1. Because the load conductivity is low enough for host cell protein to bind only weakly, much of the burden passes into the flow through and into the wash. The wash removes most of what remains, at some cost in bound antibody.

Aggregate behaves differently. On a cation exchanger, aggregate carries more charge than monomer and elutes later, which places the boundary between the two inside the descending side of the elution peak. The point at which collection stops therefore sets both how much aggregate enters the pool and how much product is recovered by the step. No other parameter of the step trades purity against yield so directly.

Across the ranges studied, protein load acts on both attributes through one mechanism: a higher load occupies more of the resin capacity, which leaves less capacity for the weakly bound impurities and moves the aggregate boundary earlier in the peak. Elution buffer pH acts instead on the strength of the interaction between antibody and resin, which sets where the whole peak sits relative to the collection window.

The same resin and buffer system are used across the Novacyte humanized IgG1 platform, on which three licensed antibodies (X-Mab, Y-Mab and Z-Mab) have been manufactured in the same elution mode. That platform history establishes the mechanisms described above and supports the ranges proposed in Table 6.1, since the separation depends on the resin chemistry and the buffer system before it depends on the molecule. Platform history does not fix the size of the effects for A-Mab, because the aggregate and host cell protein burden entering the step depends on the culture conditions (PCP-003) and on the low-pH hold (PCP-006). Therefore the size of each effect is measured in this study, with the platform contributing the mechanism and the starting ranges.

Platform experience also bounds what the step is expected to do. Charge variants and glycosylation variants are not modified by a cation exchange step operated within platform conditions, which is consistent with the binding mechanism (CMC Biotech Working Group 2009). However, the same experience gives no basis for assuming that aggregate removal is robust across the ranges now proposed, because those ranges are wider than any operated in the clinical campaigns.

4.2 Quality attributes in scope

The step forms no quality attribute. The 4 attributes in scope are formed upstream and are only reduced here. Each attribute is listed in Table 4.1 with the drug-substance criterion, the category and the criticality assigned in RA-001.

Table 4.1: Quality attributes governed by the cation exchange step.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6
Leached Protein A	process impurity	0-5 ppm	L-M	16

Aggregate is the attribute the step governs most directly, at high criticality and by the mechanism described in §4.1. Because no later step of the train reduces aggregate, the content of the pool carries through to the drug substance. The pool criterion in §7 is set on that basis.

Host cell protein is of moderate to high criticality and is cleared by three steps in sequence. Protein A removes the bulk of the burden (PCP-005), cation exchange removes a further large fraction, then anion exchange delivers the final reduction (PCP-008). Because no single step is credited with the whole clearance, §7 states the share this step must deliver.

By contrast, residual DNA is of very low criticality and clears by a wide margin across the three chromatography steps, while leached Protein A is formed at the capture step (PCP-005) and is reduced here and again at anion exchange, at low to moderate criticality.

Two of the 4 attributes are treated as responses of the designed experiments, because pool aggregate and pool host cell protein depend on the parameters in Table 6.1 through the mechanisms described in §4.1. The mechanisms in §4.1 give no basis on which residual DNA or leached Protein A would depend on those parameters. Both are therefore measured on the centre-point runs as a confirmation of platform clearance, without a model (§5.4).

Some viral clearance can be demonstrated across a cation exchange step, although no claim is made for this step. The cumulative claim for the drug substance rests on low-pH inactivation (PCP-006), anion exchange (PCP-008) and small-virus retentive filtration (PCP-009), whose contributions PCMR-001 consolidates (International Council for Harmonisation 2023a).

Step yield is a process performance attribute and carries no drug-substance criterion, but yield is measured on every run because elution stop collect trades recovery against

aggregate removal, which means a collection range chosen on aggregate alone could be unusable in manufacture.

4.3 Risk-based prioritization of parameters

Every parameter of the step was ranked in RA-001 for potential impact on the attributes in Table 4.1 and for potential interaction with the other parameters. 4 of the 5 parameters were placed in the multivariate group by that ranking, with 1 placed in the univariate group. The resulting groups and the ranges to be studied are given in Table 6.1.

The multivariate group holds protein load, load and wash conductivity, elution buffer pH and elution stop collect, because each of the four has a mechanism that links it to at least one attribute in Table 4.1 (§4.1) and a plausible interaction with at least one of the other three.

The interaction between protein load and load and wash conductivity was ranked highest, since a high load leaves less resin capacity for the weakly bound host cell protein while a high wash conductivity removes more of what remains, which makes the effect of either parameter depend on the level of the other. The interaction between protein load and elution buffer pH was ranked second, because both move the aggregate boundary relative to the collection window.

Elution flow rate was assigned to the univariate group, because flow rate acts through two mechanisms alone, residence time and peak sharpness. Neither mechanism is expected to interact with the other four parameters over the range in Table 6.1. A univariate study of that parameter is therefore sufficient, as described in §6.4.

Operating conditions that RA-001 did not rank for study, principally buffer composition, bed height and equilibration volume, are held at the platform set-points in SOP-2010. However, the study then supports the proposed ranges at those settings only, an assumption recorded in §12.

5 Materials and methods

5.1 Scale-down model and its qualification

A scale-down chromatography system will be qualified as a model of the commercial cation exchange step before any characterization run is executed. The model holds the bed height, the linear flow velocity, the protein load per litre of resin and the load, wash and elution volumes in column volumes at the commercial values. Before each block of runs, column efficiency will be verified by plate count and peak asymmetry under SOP-1001 and SOP-2008.

For the qualification, triplicate runs will be executed at the set-points in Table 6.1, on neutralized viral inactivation pool produced by a scale-down train (PCP-006), which keeps the aggregate and host cell protein burden entering the step representative of commercial operation. The performance parameters compared across scales are step yield, pool aggregate, pool host cell protein, residual DNA and leached Protein A.

The model will be accepted where no statistically significant difference is found between the scale-down and the commercial values of any of those five performance parameters at $\alpha = 0.05$, and where column efficiency meets the limits in SOP-2008. Where a difference is found, the model will be modified and requalified before the characterization runs begin. Modification and requalification of the model are required by the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), because the conclusions of the study are applied directly to the commercial process.

Runs will be executed in the temperature-controlled chromatography chamber listed in Table 5.1, which is under calibration and change control.

Table 5.1: Instrumented scale-down equipment used for this study.

Equip- ment	Description	Calibration status	Next calibration due
EQ-CHR- 118	temperature-controlled chromatography chamber	in calibration at execution	2026-05-30

Qualification supports predictions of mean performance at commercial scale, but the conclusions do not extend to feed streams outside the range of the material used for the comparison. That limit on the conclusions is recorded in §12.

5.2 Operation

Each run follows SOP-2010, under which the column is equilibrated, loaded to the target protein load, washed at the load conductivity, then eluted by a step change to the elution buffer. Collection begins on the ascending side of the ultraviolet peak at a fixed threshold and stops at the descending value given in Table 6.1. The column is then stripped, sanitized and re-equilibrated under SOP-2008.

Load conductivity and wash conductivity are prepared from the same buffer and are studied as one parameter, as Table 6.1 shows, because varying them independently would add a factor to every design without a mechanism that would justify the split. Throughout each run, temperature is held under ambient chamber control in the unit listed in Table 5.1, which keeps a variable that is not studied here from drifting between runs.

Resin lot and column will be held constant across the whole study, which removes resin lot as a source of variance between runs. The conclusions are correspondingly bounded to one resin lot. That bound is recorded in §12, while equivalence between resin lots is governed separately under SOP-2008.

5.3 Analytical methods

Every attribute in Table 4.1 is measured by a validated method, of which the four that apply to this step are listed in Table 3.2.

Size variants, including the aggregate response, are measured by size exclusion chromatography under AMV-3011, host cell protein by ELISA under AMV-3012, residual DNA by quantitative PCR under AMV-3014 and leached Protein A by ELISA under AMV-3016. For every run, step yield is calculated from the product mass in the pool and the product mass loaded, both determined by ultraviolet absorbance.

Assay variability enters the study as part of the pure error term, which the centre-point replicates of both designs separate from model lack of fit (§5.5). The largest single contributor to that pure error is the size exclusion assay, because aggregate in the pool is expected to sit low in the working range of the method. That assay risk is treated in §12.

After execution, samples will be tested inside the stability window each method validation states, with any sample that cannot be tested inside that window reported as a deviation under §9.

5.4 Sampling plan

Samples are taken twice in every run, once from the column load and once from the eluate pool. The attributes measured at each sample point, the method and the runs sampled are given in Table 5.2.

Table 5.2: Planned sampling and testing per characterization run.

Sample point	Attribute	Method	Runs sampled
Column load	Aggregate (HMW)	SEC-HPLC (AMV-3011)	every run
Column load	Host cell protein	ELISA (AMV-3012)	every run
Column load	Protein concentration	Ultraviolet absorbance	every run
Eluate pool	Aggregate (HMW)	SEC-HPLC (AMV-3011)	every run
Eluate pool	Host cell protein	ELISA (AMV-3012)	every run
Eluate pool	Protein concentration (step yield)	Ultraviolet absorbance	every run
Eluate pool	Residual DNA	qPCR (AMV-3014)	7 centre-point runs
Eluate pool	Leached Protein A	ELISA (AMV-3016)	7 centre-point runs

On every run of both designs, aggregate, host cell protein and step yield are measured, because all three are responses of the models in §6. On the 7 centre-point runs, residual DNA and leached Protein A are also measured, which confirms the platform clearance without adding runs to either design.

In addition, the load sample is measured for aggregate and host cell protein on every run, because the input level is a known source of variance at a cation exchange step (CMC Biotech Working Group 2009). Recording the input level allows an analysis adjusted for the load, where the spread of the centre points requires one. For the duration of the study, retained pool samples will be held frozen, which allows an invalidated assay result to be repeated without repeating the chromatography run.

5.5 Statistical methods

All designs are generated and analysed under SOP-1002, with effects estimated on coded factor levels, where -1 and $+1$ are the edges of the characterization range in Table 6.1. An effect is therefore the change in the response across the full range studied. Terms are judged significant at $\alpha = 0.05$.

The screening design is analysed with a model carrying the intercept, the 4 main effects and the 6 two-factor interactions, which leaves 8 residual degrees of freedom from 19 runs and 11 model terms. The response-surface design is analysed with the full quadratic model, which adds the 4 quadratic terms and leaves 13 residual degrees of freedom from 28 runs and 15 model terms.

Both model fits therefore retain enough residual degrees of freedom for a lack of fit test against pure error, which is estimated from the 3 centre-point replicates of the

screening design and the 4 replicates of the response-surface design. Neither model sits near saturation, which matters because a saturated fit cannot be tested for lack of fit at all.

The parameters that carry an effect are identified from the screening analysis, while the response-surface model is the predictive model of this study. Consequently the operating region in §7 and the proven acceptable ranges in §8 are derived from the response-surface model alone. The screening model is never presented as predictive.

Model adequacy is judged on the regression F test, the coefficient of determination with its adjusted and predicted values, the lack of fit test against pure error and the residual diagnostics, for each of which §7 states the threshold that will be applied. Within each design, the runs are executed in a randomized order, which is recorded with the data and is available to the analysis as a blocking term where a trend appears.

6 Study design

6.1 Factors, ranges and study type

The 5 parameters of the step are listed in Table 6.1 with the set-point, the range to be studied, the normal operating range and the study type. Table 6.2 gives the letter codes used in the design matrices.

Table 6.1: Parameters, set-points, characterization ranges and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	25	10-40	18-32	multivariate
Load/Wash conductivity	mS/cm	5	3-7	4-6	multivariate
Elution buffer pH	pH	6	5.8-6.2	5.85-6.15	multivariate
Elution stop collect	OD desc.	1	0.5-1.5	0.7-1.3	multivariate
Elution flow rate	cm/hr	200	100-300	150-250	univariate

Table 6.2: Factor letter codes used in the planned design matrices.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

By design, each characterization range in Table 6.1 extends beyond the normal operating range on both sides. A range that stops at the normal operating edge cannot show where performance breaks down, nor can it support a proven acceptable range wider than the one already in use.

The range studied for protein load runs from 10-40 g/L resin against its normal operating range of 18-32, the widest expansion of the five, because the upper edge covers both a campaign in which the capture pool arrives more concentrated than planned and a later increase in column productivity. Load and wash conductivity is

studied from 3–7 mS/cm against a normal operating range of 4–6, a range that covers buffer preparation tolerance and the conductivity the neutralized inactivation pool can carry into the step.

Elution buffer pH is studied from 5.8–6.2 against a normal operating range of 5.85–6.15, which is the narrowest expansion of the five, since a wider range would move the elution outside the platform window and would test a step the process would never run. However, the narrow range also limits what the response-surface model can say about pH, a limit recorded in §12.

The range for elution stop collect runs from 0.5–1.5 OD on the descending side of the peak, against a normal operating range of 0.7–1.3, where the lower edge is a conservative collection that gives up yield for purity and the upper edge collects into the aggregate boundary described in §4.1. For elution flow rate the range runs from 100–300 cm/hr against a normal operating range of 150–250, with the upper edge bounded by the pressure the packed bed tolerates under SOP-2008.

6.2 Screening design

The 4 multivariate parameters enter a two-level factorial design of 16 factorial runs and 3 centre-point replicates, 19 runs in total. The design is full in the four factors. Every main effect and every two-factor interaction is therefore estimable without aliasing, as the interaction ranking in §4.3 requires. The planned design matrix is given in Appendix A, in coded factor levels. The purpose of that design is to identify the parameters whose effect on the three responses is largest, after which the parameters identified are carried into the response-surface stage (U.S. Food and Drug Administration 2011). A screening design of this size does not estimate curvature, because every factorial point sits at an edge of the range. The centre-point replicates indicate whether curvature is present, without locating it in any one factor.

Although Appendix B is written on the assumption that all four multivariate parameters carry an effect, the screening stage may identify fewer, in which case the matrix will be re-issued before execution as an amendment to this plan.

6.3 Response-surface design

The parameters carried forward from screening enter a face-centred central composite design of 16 factorial runs, 8 axial runs and 4 centre-point replicates, 28 runs in total. The axial points sit on the faces of the coded cube at ± 1 , which keeps every run inside the ranges in Table 6.1. The planned matrix in coded factor levels is given in Appendix B.

At the response-surface stage, the fitted model carries the intercept, the 4 main effects, the 6 two-factor interactions and the 4 quadratic terms, with each quadratic term estimated from three levels of the factor. Three levels estimate curvature less precisely than a rotatable design would, which is a real cost of the choice. However, a rotatable

design places the axial runs outside the coded cube, where they would fall outside the characterization ranges this plan claims. The faced design was chosen for that reason.

6.4 Univariate assessment

Elution flow rate is assessed one parameter at a time, at 3 levels: the low edge of the characterization range, the set-point and the high edge (Table 6.1). While flow rate moves, the other four parameters are held at their set-points in Table 6.1. Each level is executed in duplicate, which gives 6 runs.

The three levels bracket the normal operating range, with the same three responses measured as in the multivariate designs and under the same methods (Table 5.2).

A univariate design cannot determine an interaction between flow rate and the other parameters, where the effect of flow rate would depend on the level of the others. That design therefore supports a proven acceptable range for flow rate at the set-points of the other four parameters only. The limit on that range is recorded in §12. Together with the two multivariate designs, the univariate block brings the study to 53 characterization runs.

7 Acceptance and decision criteria

Two sets of criteria apply to this step in different places, since the drug-substance criteria in Table 4.1 apply to the final drug substance after every remaining step of the train, while the criteria in Table 7.1 apply to the cation exchange pool itself. Every decision rule in this plan is taken against the pool criteria, which are the criteria this step alone can be held to.

Table 7.1: Acceptance criteria applied to the cation exchange pool, with the drug-substance criteria for reference.

Attribute	Criticality	Drug-substance criterion	Pool criterion	Basis
Aggregates (HMW)	H	0-5 % HMW (SEC)	≤ 1	in-process
Host Cell Protein (HCP)	M-H	0-100 ng/mg	≤ 193	in-process
Residual DNA	VL	0-0.001 ng/dose	—	confirmation only
Leached Protein A	L-M	0-5 ppm	—	confirmation only

The pool is an intermediate and cannot be judged against a drug-substance criterion where later steps still reduce the attribute, as anion exchange does for host cell protein after this step (PCP-008). The pool criterion for host cell protein is therefore the drug-substance criterion of 100 ng/mg carried back through the clearance credited to anion exchange, divided by an assurance margin of 3.2. The resulting pool criterion is 193 ng/mg. That criterion works as an in-process control instead of a break-even point, because the margin holds it above the value at which the drug substance would depend on every later step delivering nominal clearance exactly.

The aggregate criterion is derived differently, because no later step of the train reduces the level once the pool has been collected. The back calculation for aggregate therefore carries the drug-substance criterion of 5 % HMW through unchanged, which leaves the assurance margin of 5 to account for the whole of the difference. That calculation gives a pool criterion of 1 % HMW.

Residual DNA and leached Protein A carry no pool criterion in Table 7.1 and enter no decision rule of this plan, being measured against the drug-substance criteria as a confirmation of platform clearance (§5.4).

A response-surface model will be accepted as predictive for a response only where four conditions hold together. The regression of the model is significant at $\alpha = 0.05$,

the lack of fit test against pure error is not significant at $\alpha = 0.05$, the predicted coefficient of determination stays close to the adjusted value by the numerical rule in SOP-1002, with the residual diagnostics showing no pattern against predicted value or against run order.

Where a model fails any of the four conditions, the response will be reported without a design space claim. The parameters that govern that response will then be held at their normal operating ranges in Table 6.1. That response and the reason for the failure will be reported in PCR-007.

The operating region will be declared as the part of the characterized space in which the mean prediction of every accepted model meets the pool criterion in Table 7.1, bounded by the ranges in Table 6.1 and by the scale-down assumptions in §5.1. A mean level model gives roughly even odds of meeting a criterion at the edge of the region, where the variability is symmetric about the mean (CMC Biotech Working Group 2009). For this reason the normal operating ranges in Table 6.1 must lie inside the declared region. The same ranges are re-examined in §8 by an analysis that propagates the variation through the model.

Once the analysis is complete, each parameter will be classified under SOP-4001 on the evidence the study produces. A parameter whose variation across the characterization range moves a governed attribute towards the pool criterion will be classified as quality linked. A parameter with no resolvable effect on any governed attribute will be classified as a general process parameter and will be retained in the knowledge space as evidence of robustness. Each parameter classification will be reported in PCR-007 with the reason for it.

The study passes where three conditions hold at the end of execution: the scale-down model is qualified under §5.1, every response either has an accepted model or has a stated reason for the absence of one, with the declared operating region containing the normal operating ranges of all 5 parameters. A failure of any of the three is itself a study outcome and will be reported as such in PCR-007, together with the ranges the data do support.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of one parameter over which a governed attribute stays within acceptance, with the other parameters at stated settings (International Council for Harmonisation 2009). That range will be determined for each of the 2 governed attributes against each of the 4 response-surface parameters, which gives 8 combinations. Each combination is judged against the pool criterion in Table 7.1.

Each combination is covered by two analyses. The first analysis holds the other parameters at their set-points and evaluates the fitted response-surface model across the characterization range of the parameter of interest, on a grid of 201 points. The second analysis varies the other parameters within their normal operating ranges in Table 6.1 and propagates that variation through the same model by Monte-Carlo, on a grid of 81 points.

Each Monte-Carlo grid point draws 2,000 values, in which settings for the other parameters are taken from a normal distribution centred on the set-point, with the normal operating range spanning six standard deviations and each draw clipped to the characterization range. A residual term with the standard deviation of the fitted model is then added to each draw. Each draw therefore carries model error as well as parameter variation. The decision rule differs between the two analyses. Under the first, the parameter range qualifies where the mean prediction meets the pool criterion, while under the second the range qualifies where the 95 % predictive interval meets the same criterion. The second rule is a robustness rule and returns the narrower of the two ranges.

The reported range is the contiguous interval around the set-point over which the rule holds. Where the set-point itself fails the rule, no proven acceptable range is reported for that combination. PCR-007 will state that outcome plainly.

The set-point of each of the 4 response-surface parameters lies at the midpoint of the characterization range in Table 6.1, which means the first analysis evaluates the model at the centre of the coded design as well as at the set-points.

Each parameter will be reported against the attribute as a plot, with the acceptable region shaded green and with the set-point and the normal operating range marked. One such plot will be carried in PCR-007 for each governed attribute, against the parameter with the largest main effect on that attribute. The remaining combinations will be reported in tabulated form.

The analysis rests entirely on the fitted response-surface model, which bounds every range it returns. Therefore no proven acceptable range will extend beyond the

characterization range in Table 6.1, nor beyond the scale-down assumptions of §5.1 and the model assumptions of §7.

9 Data management and integrity

Throughout the study, data are recorded and retained under the Novacyte data integrity procedures, with raw chromatograms, run records and assay results captured in the validated systems that own them and with audit trails enabled. Records follow ALCOA+, which requires each record to be attributable, legible, contemporaneous, original and accurate, as well as complete, consistent, enduring and available.

Before analysis, every design matrix, every executed run order and every dataset is reviewed by a second person. The analysis itself is reviewed by Statistics under SOP-1002.

During execution, deviations from this plan are recorded as they occur and are assessed for impact on the affected design, with PCR-007 stating the disposition of each one. A run excluded from an analysis will be reported with the reason for exclusion, which must be established before the analysis has been seen.

10 Roles and responsibilities

The functions listed in Table 10.1 execute this plan, with Process Development owning the plan, the execution and the report.

Table 10.1: Functions and responsibilities for the execution of this plan.

Function	Responsibility
Process Development / MSAT	Owens this plan, executes the runs, writes PCR-007
Statistics	Generates the designs, performs the analysis, judges model adequacy
Analytical Development	Runs the validated methods, reports assay performance
Manufacturing Science	Supplies commercial-scale data for the scale-down comparison
Quality Assurance	Approves this plan, oversees deviations, approves PCR-007

Quality Assurance approves this plan before execution begins and approves PCR-007 on completion. Statistics is independent of execution, which keeps the analysis separate from the group that generates the data.

11 Deliverables and schedule

The deliverable of this plan is PCR-007, the process characterization report for the cation exchange step, which will report the qualification of the scale-down model, the screening result, the response-surface models, the operating region, the proven acceptable ranges and the parameter classification. PCR-007 in turn rolls up into the master report for the campaign (PCMR-001), which consolidates the operating regions and the classifications of every step.

The work runs in four phases in sequence, beginning with scale-down model qualification, followed by the screening design of 19 runs, then the response-surface design of 28 runs, and completed by the univariate assessment of 6 runs. The response-surface stage does not begin until the screening result has been reviewed under §6.2. The univariate runs may be executed in parallel with either multivariate block, because the analysis of those runs is independent of both.

No calendar dates are fixed in this plan, because the campaign schedule is held in the transfer plan (PTP-001) and is maintained there.

12 Risks and assumptions

The largest risk to this study is the representativeness of the scale-down model, because a chromatography model that matches bed height, linear velocity and load can still differ from the commercial column in wall effects and in flow distribution. Qualification under §5.1 addresses the risk, since a failure at that stage stops the study instead of biasing the result.

The second risk is the supply of representative load material, since every run needs neutralized viral inactivation pool with an aggregate and host cell protein burden typical of commercial operation. Material outside that burden would shift the level of the responses without necessarily changing the effects, which is the risk the load measurement on every run addresses (§5.4). The third risk is analytical, in that aggregate at the level expected in the pool sits low in the working range of the size exclusion method. Assay variability then contributes a large share of the pure error, which the centre-point replicates will quantify.

Four assumptions underlie this plan. The first of those assumptions treats the risk ranking in RA-001 as correct, which means a parameter ranked low there is held at set-point and is not studied here. The characterization ranges in Table 6.1 are treated as wide enough to expose any edge of failure that matters to the control strategy. One resin lot is treated as representative of the resin used at commercial scale (§5.2). The quadratic model form is treated as adequate over the ranges studied, an assumption that the lack of fit test in §7 exists to test.

Each assumption is recorded here so that PCR-007 can report against it. An assumption that fails during execution becomes a deviation under §9.

13 Approvals

This plan takes effect when the approvers named in the approval record at the front of the document have signed it. The signatories are listed there by role. No characterization run may be executed before that record is complete.

Appendix A — Planned screening design matrix

The coded matrix of the planned screening design is given in Table 13.1. Factor letters are defined in Table 6.2. The natural value of each coded level follows from the characterization ranges in Table 6.1. Response columns are absent because no run has been executed.

Table 13.1: Planned screening design, coded factor levels.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Appendix B — Planned response-surface design matrix

The coded matrix of the planned response-surface design is given in Table 13.2, on the same convention as Table 13.1. The matrix is written on the assumption stated in §6.2 and will be re-issued if the screening stage identifies fewer parameters.

Table 13.2: Planned response-surface design, coded factor levels.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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